HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F70LA

Life Insurance Mathematics A

Semester 1 2019/20

Duration: Two Hours

Please answer all 6 questions.

Total marks: 60

Electronic calculators may be used, provided that they conform to University regulations.

You should have a copy of the yellow Formulae and Tables for Actuarial Examinations on your desk.

Question 1

Use AM92 in calculating the following. Write your answers correct to six decimal places.

(a) ${}_3p_{40}$ [2 marks]

(b) ${}_{9|5}q_{40}$ [2 marks]

(c) ${}_6q_{[40]}$ [3 marks]

(d) ${}_{0.3}p_{[40]}$ assuming a uniform distribution of deaths between integer ages. [3 marks]

(e) ${}_{3.8}q_{40}$ assuming constant force of mortality between integer ages. [4 marks]

(f) When an actuary is choosing between a uniform distribution of deaths and a constant force of mortality between integer ages, how important is it that she makes the right choice? [1 mark]

[Total 15 marks]

Question 2

State, with reasons, whether or not each of the following functions is a well-defined survival function:

(a) $S_0(t) = (50-t) / 50$ $(0 \le t \le 50)$ [2 marks]

(b) $S_0(t) = 2\exp(-4t) - 1$ $(0 \le t < \infty)$ [2 marks]

(c) $S_0(t) = (900 + 80t - t^2) / 900$ $(0 \le t \le 70)$ [3 marks]

[Total 7 marks]

[PLEASE TURN OVER]

Question 3

(a) Write down the symbol for the expected present value of a term assurance for a life aged 40, with term 20 years, on select mortality. The benefit, of £1, is paid immediately on death. [1 mark]

(b) Calculate the value of the symbol in (a), assuming AM92 mortality select, and an interest rate of 6%. [4 marks]

[Total 5 marks]

Question 4

Calculate the expected present value as at 1 January 2020 of a special immediate annuity of £1 per year paid continuously for life, to a woman aged 65 on 1 January 2020. If she dies before 1 January 2025, the payments will continue to be paid until 1 January 2025. The basis is: PFA92C20, 4% interest.

[Total 4 marks]

[PLEASE TURN OVER]

Question 5

A man now aged 50 took out an endowment contract at the age of 20. The contract promises him £1,000,000 at age 65 or at the end of the year of death, whichever is sooner.

The contract states that he must pay premiums of £4,000 per year, in advance, for the whole of the term of the contract.

(a) He now wishes to change the contract to a whole-life contract, with premiums remaining at £4,000 per year for the rest of his life, and a different sum assured. Calculate the new sum assured.

The basis for the change is as follows.
Mortality: AM92 ultimate
Interest: 4%
Alteration expense: £500
No other expenses. [6 marks]

(b) A student says that the original premium was too low, and that this should be allowed for when the new sum assured is calculated. Comment on this suggestion. [2 marks]

(c) Another student says that there are selection issues in the proposed change. Comment on this suggestion. [3 marks]

[Total 11 marks]

[PLEASE TURN OVER]

Question 6

Years ago, a life office sold a number of deferred annuities to men born on 1 January 1954. Each of these annuities pays an annuity of £10,000 annually in advance, starting at age 65. If a policyholder dies before age 65, a lump sum of £50,000 is paid at the end of the year of death.

At 1 January 2018, 80 of these contracts were in force. The life office uses the following basis for valuations.

Mortality: PMA92C20
Interest: 4%
Expenses for the whole block of business: £20,000 per year paid continuously until 1 January 2038, then nothing.

(a) Calculate the value of these contracts on the above basis. [6 marks]

(b) In 2018, the experience of the life office was as follows.
One death.
Interest 2.2%.
Expenses: £30,000 paid at 31 December 2018

Calculate the profit or loss to the office during 2018, as at 31 December 2018. [4 marks]

(c) How much of this profit or loss arises from each of the three basis elements: mortality, interest and expenses? [4 marks]

(d) The student from question 5(b) tells a friend at another company that the actuary's method of valuing expenses is nonsense. Discuss this suggestion. [4 marks]

[Total 18 marks]

END OF PAPER